

Smart Vault Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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Immeas

September 13, 2024

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for The Standard DAO on Smart Vault. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated September 13, 2024, was delivered to The Standard DAO, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 8 days, the Cyfrin team conducted an audit on the Smart Vault code provided by The Standard DAO. The findings consist of 2 Critical, 2 High, 3 Medium & 8 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Smart Vault
Repository	smart-vault
Commit	c6837d4a296f...
Audit Timeline	Aug 26th - Sep 4th, 2024
Methods	Manual Review, Stateful Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	2	2	0
High Risk	2	2	0
Medium Risk	3	3	0
Low Risk	8	6	2
Informational	5	4	1
Gas Optimizations	2	2	0
Total	22	19	3

6 Audit Scope

Cyfrin conducted an audit of The Standard Smart Vault based on the code present in the repository commit hash c6837d4.

The following were included in the scope of the audit:

```
contracts/SmartVaultV4.sol#L79-L103
contracts/SmartVaultV4.sol#L279-L338
contracts/SmartVaultYieldManager.sol
```

SmartVaultYieldManager.sol was taken from the usds branch.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The Standard is an overcollateralized lending protocol and issuer of the USDs stablecoin. Smart Vault owners can borrow USDs against their deposited collateral, swap directly between accepted collateral tokens, and deposit their collateral into selected Gamma Vaults (aka Hypervisors) for additional yield-earning opportunities.

9 Attestation

Cyfrin attests that it performed a security review of the Smart Vault code provided by The Standard DAO.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-09-13-cyfrin-the-standard-smart-vault-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to The Standard DAO has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=e2bb9c7730f2c459528507db99faed6e1dfa172cf329e79744e6d8ef5428d5ed
F=2024-09-13-cyfrin-the-standard-smart-vault-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.